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Anthropomorphizing AI: Perceptions, Interpretations, and the Humanization of Generative AI

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Abstract

The rise of generative artificial intelligence (AI), particularly advanced AI chatbots such as ChatGPT, has transformed human–AI interactions by providing unprecedentedly natural interaction environments. These developments have significantly heightened user perceptions of AI as human-like, making anthropomorphism, a tendency to attribute human traits to non-humans, increasingly prevalent. Despite the recognition of the rising anthropomorphism in these interaction modalities, little is known about how people anthropomorphize generative AI, especially in social media discourse. Using Twitter/X as a discursive space, we systematically analyze how users described and engaged with ChatGPT, identifying distinct categories and recurring forms of anthropomorphism. Findings reveal diverse anthropomorphic perceptions, including personification, emotional attributions, cognitive and human-like attributes, social interactions, and human-like personality traits. Building on the observation that social media platforms not only reflect but also actively shape and reinforce anthropomorphic narratives, this study raises important implications for trust, misinformation, user expectations, and human–AI interactions.

Keywords

ChatGPT, generative AI, anthropomorphism, Twitter/X

Introduction

The proliferation of generative artificial intelligence (AI) represents a key shift in human–computer interaction (HCI). In contrast to earlier technologies, generative AI systems offer advanced conversational abilities, characterized by unparalleled contextual awareness and conversational fluency (Maeda & Quan-Haase, 2024; Rafikova & Voronin, 2025). These capabilities significantly amplify user perceptions of human-like intelligence and actions, potentially fostering greater anthropomorphism among users (Akbulut et al., 2024; Alabed et al., 2022; DeVrio et al., 2025; Polyportis & Pahos, 2025). Anthropomorphism can be broadly defined as the attribution of distinctly human characteristics, such as conscious thought, emotions, motivations, and intentions, to non-human entities, including robots or AI systems (Epley et al., 2007).

Studies indicate that the perception of anthropomorphism among users of large language models (LLMs) has rapidly increased (Chaturvedi et al., 2025). Recent large-scale surveys indicate a striking prevalence of anthropomorphic interactions, with 38% of users anticipating the formation of deep human-like relationships with LLMs within the next decade (Rainie, 2025). Such perceptions are not incidental but are often intentionally encouraged by AI design through specific linguistic markers and behaviors, which actively promote anthropomorphic interpretations (DeVrio et al., 2025; Reinecke et al., 2025; Sherman et al., 2025). For example, some LLMs such as Anthropic's Claude have human names and many use specific human strategies such as politeness (Ribino, 2023). Despite warnings from researchers and even AI developers themselves urging the public to treat AI as tools rather than a sentient actor (Desai & Twidale, 2023), the intentional system design choices, such as the use of first-person pronouns,

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emotional expressions, and relationship-building cues, may still contribute to perceptions of anthropomorphism (e.g., AI as possessing personal agency, emotions, and self-awareness) (Mitchell, 2024).

Understanding such anthropomorphic tendencies in generative AI and their impact on human–AI interactions is crucial due to their significant psychological and societal implications (Akbulut et al., 2024; Cohn et al., 2024; DeVrio et al., 2025). For example, users who attribute emotions, understanding and free will to intelligent systems may experience reduced personal agency, moral confusion about responsibility, overestimations of system capabilities and unwarranted trust in potentially incorrect or biased outputs (Abercrombie et al., 2023; Friedman & Kahn, 2007; Inie et al., 2024). Conversely, some research suggests that anthropomorphizing algorithms may enhance the effectiveness and interactions with systems, as they simulate natural interactions (Bankins & Formosa, 2020; Seeger et al., 2021). Beyond these normative assessments of anthropomorphism and its impact across social contexts, understanding how it manifests in emerging interactions with generative AI is a significant research undertaking as these systems increasingly pervade the information landscape of hundreds of millions of users and play a crucial role in human decision-making (Maeda & Quan-Haase, 2024; Obrenovic et al., 2025). Beyond purely functional uses, human–machine communication (HMC) scholarship stresses the relational and metaphysical aspects of AI (Guzman & Lewis, 2020), including how such technologies are understood, framed, and discursively constructed (Inie et al., 2024; R. Zhang, Li, et al., 2024), calling for broadened perspectives that transcend individual-centric questions and concerns to societal ones (Hepp et al., 2023).

In this context, social media platforms provide a valuable setting for examining how public discourse collectively constructs and reinforces anthropomorphic perceptions of generative AI (Demirel et al., 2025). In contrast to user interactions in controlled settings, social media interactions and their digital traces offer insights into how users collectively and organically make sense of AI identity, characteristics, and roles (Karami et al., 2024). Despite the substantial literature on anthropomorphism in psychology and HCI, we know less about how anthropomorphism is expressed in the wild within large-scale, public discourse about generative AI. Prior work has theorized why people anthropomorphize (e.g., the sociality, effectance, and elicited knowledge [SEEK] model; Epley et al., 2007) and empirically investigated what the consequences of anthropomorphism are (Waytz et al., 2010). It has also cataloged design and linguistic cues that can elicit anthropomorphic attributions (e.g., Akbulut et al., 2024; DeVrio et al., 2025; Inie et al., 2024). We build on these foundations by offering a bottom-up, inductive typology of how users anthropomorphize ChatGPT on Twitter/X, answering the following research question:

Research Question: What forms of anthropomorphism are expressed in public Twitter/X conversations about ChatGPT?

Our goal is descriptive and classificatory, namely to delineate the types of anthropomorphism that occur in public posts and to provide representative examples, thereby complementing feature-based and experimental approaches with an account of naturally occurring, socially shared framings.

Literature Review

Anthropomorphism

Over the past few decades, anthropomorphism has been an enduring subject of research across disciplines such as psychology, philosophy, religious studies, and cognitive science (Daston & Mitman, 2005; Epley et al., 2007; Guthrie, 1993). Broadly construed, anthropomorphism refers to the attribution of human-like qualities or behaviors to non-human entities. It is observed in both children and adults (Seeger et al., 2021). Epley et al. (2007), in a seminal contribution and synthesis, defined anthropomorphism as the tendency of "attributing humanlike properties, characteristics, or mental states to real or imagined nonhuman agents and objects" (p. 865). More concisely and recently, Epley (2018) described anthropomorphism as "simply perceiving human-like traits in nonhuman agents" (p. 591). This involves assigning human mental states and emotions to non-humans, such as animals, robots, or even inanimate objects, and includes both behavioral (e.g., speaking to a cuddly toy as if it were a human friend) and attitudinal patterns (e.g., assuming a malfunctioning computer is angry or refuses to cooperate). The psychological and social processes contributing to anthropomorphism can be described with the SEEK model, where SEEK motivations together affect anthropomorphic attitudes and behaviors in dispositional, situational, developmental, and cultural ways. For example, sociality motivations can include chronic loneliness as a disposition, situational disconnection (e.g., arriving in a new country where one has limited social connections), developmental patterns (e.g., insecure attachment from childhood), and being part of a collectivist culture that is more prone to animism—all of which are expected to increase anthropomorphism.

Research has shown that anthropomorphism is not an isolated perceptual or descriptive act, but is also deeply influenced by human motivations and contextual cues (Lombard & Xu, 2021). For example, when people face uncertainty, isolation, or unpredictability, they are more likely to anthropomorphize (Epley et al., 2007; Waytz et al., 2010). Scholars have also distinguished anthropomorphism from related concepts, such as personification, animism, and mentalizing. They have highlighted its role in how individuals interpret and emotionally relate to non-human agents (Złotowski et al., 2017).

Increasingly, scholars view anthropomorphism not only as a cognitive bias but also as a communicative and affective strategy that serves functional purposes in human interaction with systems and environments (Złotowski et al., 2017). It is common for people to anthropomorphize religious figures, animals, the environment, and technological artifacts, ranging from robots to intangible computational systems (Salles et al., 2020). This tendency is particularly relevant in the context of human–AI interactions, where anthropomorphizing AI systems can influence user behaviors and encourage social connections and adherence to social norms (Duffy, 2003; Maeda & Quan-Haase, 2024).

Anthropomorphism in Human–Computer Interaction

Given the growing prominence of HMC in contemporary life, research in fields such as HCI and media studies has increasingly focused on anthropomorphism in relation to intelligent systems. Notably, much of the early research focused on people's interactions with robots, which are understood as embodied, intelligent systems (e.g., Duffy, 2003; Vidal, 2007). The mere physical presence of a robot can foster human engagement and anthropomorphism, especially when robots have human-like features, such as legs instead of wheels or advanced sensory capabilities (Lombard & Xu, 2021; Złotowski et al., 2017). Other characteristics, such as (human-like) movement, also affect the anthropomorphism of robots (Mori et al., 2012). The perception that a robot moves at a similar speed to humans increases mind attribution, supported by neurological research showing that humans' social information processing center is activated by body, mouth, and hand movements (Kühne & Peter, 2023). However, the discussion of anthropomorphism has extended beyond physically embodied robots that exhibit human-like physical characteristics (Araujo, 2018).

Previous studies have demonstrated that people are particularly likely to attribute human-like qualities to software systems when they exhibit social behaviors (Reeves & Nass, 1996). The increasing use of anthropomorphic cues in system interfaces, such as language style, gendered voices, and perceived personality traits, can similarly trigger anthropomorphism among users (DeVrio et al., 2025; Kim & Sundar, 2012). Akbulut et al. (2024) trace the psychological, social, and ethical implications of anthropomorphic design across social robots, voice assistants, and generative AI. They argue that increasingly human-like interfaces foster overreliance, misplaced trust. and emotional attachment, which can heighten privacy risks, dependency and disillusionment as boundaries between humans and machines increasingly blur. Their framework calls for anticipatory governance and calibrated design to address these emerging risks. Building on this perspective, Inie et al. (2024) and DeVrio et al. (2025) shift attention from design features and user psychology to the linguistic and discursive dimensions of anthropomorphism, examining how language (both in system outputs and system descriptions) shapes perceptions of humanness and agency in AI.

The tendency for anthropomorphism is particularly visible in AI-powered voice assistants (e.g., Alexa and Siri; see Wagner et al., 2019), which users are more likely to anthropomorphize when there are politeness markers (e.g., "please," "thank you") or human-like vocal tones (Purington et al., 2017). More advanced interfaces provide drivers such as linguistic fluency, perceived intentionality, and emotional expressions that actively contribute to anthropomorphism (Akbulut et al., 2024; DeVrio et al., 2025; Seeger et al., 2021). For example, linguistic fluency is the AI's ability to facilitate a natural interaction with users by generating coherent, context-aware responses, while perceived intentionality creates the perception that AI has a motive for its actions by acting as if it is deciding or thinking on its own. Finally, emotional expressions that mimic human emotions, such as apologetic phrases, encourage users to develop an anthropomorphic connection with AI (Mallick et al., 2024).

Anthropomorphizing Generative AI

Generative AI chatbots have introduced a unique dynamic in anthropomorphism by providing unprecedented natural modes of user interaction (Akbulut et al., 2024). This is a result of their intentionally designed system, which might encourage users to anthropomorphize AI (Cohn et al., 2024; DeVrio et al., 2025; Inie et al., 2024). Since its inception, the Turing Test has served as a guiding concept and benchmark for designing and evaluating intelligent systems. The core idea is that a machine can be judged as possessing human-like thinking abilities if it can engage in a conversation with a human to the extent that it is indistinguishable from a real person (Xie, 2024). This design ideal has influenced the design process for AI chatbots, such as ChatGPT, to intentionally mimic and replicate human-like social interactions (Cohn et al., 2024). Thus, the design of these systems embodies what researchers refer to as anthropomorphic behaviors (Alabed et al., 2022; Karami et al., 2025). For example, the expression of personality through humor and creativity creates the perception of generative AI possessing and presenting a distinct human-like personality (DeVrio et al., 2025; J. Zhang, Wang, et al., 2024). These systems are also increasingly designed to use first-person pronouns and display relationship-building behaviors, such as showing empathy and validation, to foster social connections with AI (Akbulut et al., 2024; Ibrahim et al., 2025). Emerging research has also focused on the implications of generative AI anthropomorphism, discussing both benefits such as enhanced intrinsic motivation and engagement in educational settings (Reinecke et al., 2025), and risks such as overreliance, privacy, risks, seduction,

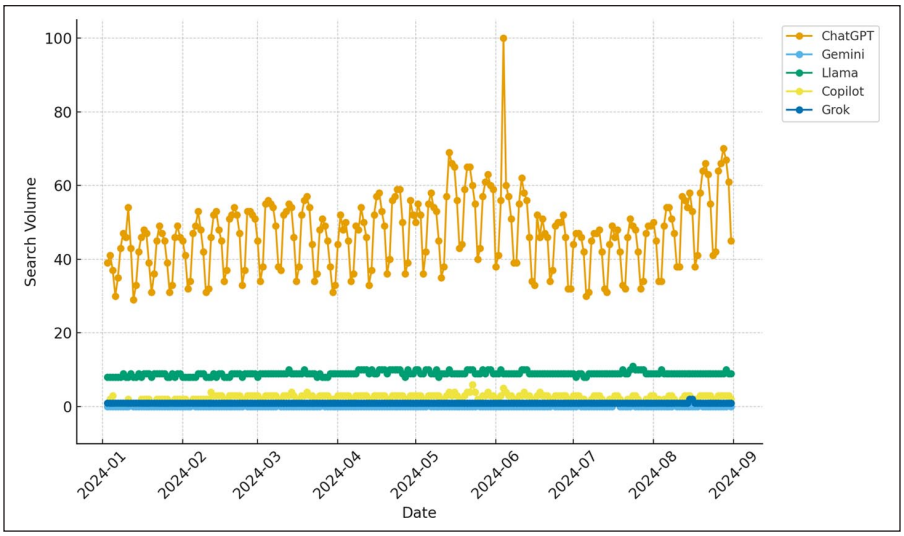


Figure 1. Google search trend for ChatGPT, Gemini, Llama, Copilot, and Grok between January and August 2024.

and deception (Akbulut et al., 2024; DeVrio et al., 2025; Lutz et al., 2026; Peter et al., 2025).

While extensive research has focused on how users anthropomorphize robots or voice assistants, research related to generative AI anthropomorphism is in its early stages (Ji, 2024; Karami et al., 2025; Kumar et al., 2025). This raises the question of how certain types of AI are anthropomorphized without the aid of commonly recognized factors, such as physical form and vocal presence. Most of the current research on anthropomorphism of generative AI relies on experiments or surveys (e.g., Alabed et al., 2022; Inie et al., 2024; R. Zhang, et al., 2025). These studies are limited as they examine controlled interactions and do not capture naturally occurring, large-scale conversations. As a result, we do not fully understand how users collectively anthropomorphize and interpret AI's capabilities in real-world settings. Moreover, while helpful, recent typologies of AI anthropomorphism (DeVrio et al., 2025) and anthropomorphization (Inie et al., 2024) primarily focus on the developer side (i.e., how AI technologies are designed to resemble humans or marketed to invoke a human-like perception or how AI-generated text intentionally embeds clues and signals), neglecting how users may represent anthropomorphism in their own interpretations and descriptions of AI. Finally, while some research explores individual user perceptions of chatbots, it rarely considers the influence of networked digital conversations on platforms such as Twitter/X and how ChatGPT's perceived identity is shaped and redefined. Recent research has looked at social media conversations about ChatGPT in general (Cohen et al., 2026; Demirel et al., 2025; Xu et al., 2024) but has not done so specifically from an anthropomorphism perspective. We have not found any publication that compares the forms of anthropomorphism attributed to ChatGPT and how it is perceived as a variety of personas, such as a helpful friend, an authoritative expert, or even an unreliable trickster. This study aims to address these gaps by examining user conversations on Twitter/X and providing insight into how ChatGPT is anthropomorphized.

Method

We employed a qualitative content analysis of public discourse on Twitter/X to investigate how users anthropomorphize ChatGPT. Twitter/X was selected as the primary site of analysis due to its role as a dynamic and participatory platform where users openly share experiences, opinions, and interpretations of technological systems (R. Wang et al., 2024). We performed a *t* test to compare Google Trends data for ChatGPT, Microsoft Copilot, Google Gemini, Meta Llama, and xAI Grok between January and August 2024. The analysis revealed that searches for ChatGPT were significantly higher ($p < .05$) than those for the other chatbots, showing its greater popularity during this period. Figure 1 presents the normalized Google search trends on a 0–100 scale, where a value of 100 represents the term's peak popularity within the selected timeframe, and a value of 50 indicates half that level of interest.

More recent evidence confirms that ChatGPT is still by far the most used generative AI system (Data Studios, 2025). As a discursive space, Twitter/X enables the public negotiation of AI's identity through spontaneous, large-scale, and culturally embedded conversations (Lewis et al., 2025).

Data Collection

To construct a focused and relevant dataset, we used Twitter/X's Advanced Search API to collect English-language tweets that mentioned "ChatGPT." The dataset included 5,137 tweets posted by 4,708 unique users from an 8-month period between January and August 2024, chosen to capture

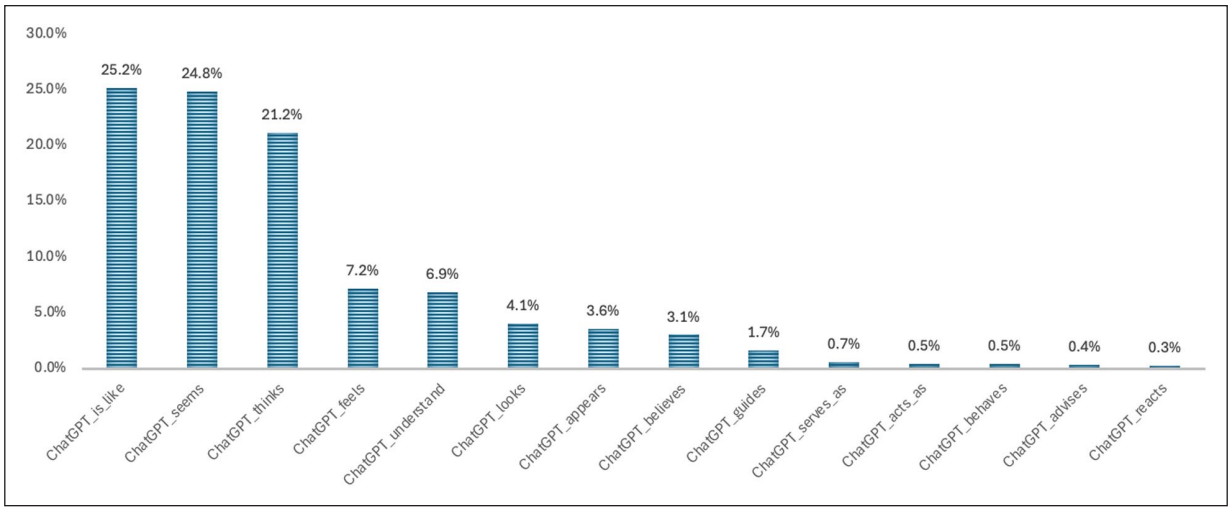


Figure 2. Percentage of queries.

a timeframe when public engagement with ChatGPT was active and evolving. This specific timeframe was chosen to capture a phase of intensified public engagement and evolving discourse around ChatGPT, following a series of major developments in generative AI (Singla et al., 2025). January 2024 marked the point when ChatGPT achieved sustained mainstream adoption, which renewed public attention and comparison across platforms (Bick et al., 2024). Meanwhile, the months leading up to August 2024 encompassed multiple milestones, including the release of two GPT-4 updates (Wikipedia, n.d.), contributing to increasing media coverage, shifts in user perception, and interaction patterns. The timeframe reflects a mature yet evolving stage in public engagement, allowing for a focused exploration of how user interest and sentiment evolved as generative AI tools transitioned from novelty to widespread integration. To ensure that the data reflected original discourse, retweets, quotes, and duplicates were excluded. Our filtering aimed to retain tweets that contained substantive commentary or descriptions of interaction with ChatGPT. In our dataset, users posted between one and 12 tweets, with an average of 1.1 tweets per user. The vast majority (93.5%) posted only one tweet, 5.4% posted two tweets, and approximately 1% contributed three or more tweets.

An iterative filtering process was then applied to isolate tweets that explicitly described ChatGPT's behavior, capabilities, or personality traits, particularly those that signaled anthropomorphic framing. Drawing on both a preliminary reading of a subsample of 500 tweets and prior research on anthropomorphism as discussed above, we identified a list of keywords and phrase structures commonly used to ascribe human-like qualities to non-human entities to ensure the inclusion of tweets containing narrative or evaluative accounts of user-AI interactions. Our query was as follows: "ChatGPT is like" OR "ChatGPT seems" OR "ChatGPT feels" OR "ChatGPT appears" OR "ChatGPT looks" OR "ChatGPT behaves" OR "ChatGPT reacts" OR "ChatGPT understand" OR "ChatGPT believes" OR "ChatGPT advises" OR "ChatGPT guides" OR "ChatGPT acts as" OR "ChatGPT serves as" OR "ChatGPT thinks." Figure 2 presents the percentage distribution of query types in our dataset, with "ChatGPT is like," "ChatGPT seems," and "ChatGPT thinks" emerging as the three most frequent queries.

We acknowledge that our unique sampling strategy embodies some trade-offs. Terms such as "ChatGPT behaves" and "ChatGPT reacts" represent an ambiguous semantic space since they can pertain to the technical functionality of AI systems as well as users' anthropomorphic interpretation of them. However, we consider this ambiguity generative for this research, as we used the first step as a sampling heuristic rather than a classification approach. Our keyword-based sampling drew on the anthropomorphism literature (DeVrio et al., 2025; Epley et al., 2007), which identifies verbs as markers exhibiting agency attribution. It was designed not to capture all instances of anthropomorphic content exhaustively, but rather to surface tweets with high potential for anthropomorphic framing (i.e., posts where users linguistically positioned ChatGPT as a social or intentional actor, allowing for deeper interpretive analysis). This sampling approach resulted in a corpus rich enough to warrant deeper manual scrutiny. As such, we deliberately cast a wide net to capture a sample space in which anthropomorphic instances were more likely to be evident.

Through manual review, we then focused on tweets that explicitly entailed human-like states/capacities or that employed linguistic personification of the system as a social actor. We recognize that such an approach comes with a limitation, as it could have missed anthropomorphic discourse embedded in other linguistic patterns, and that our focus on explicit verb phrases may have been constrained to only some observable forms of anthropomorphism. However, our methodological design, which began with keywords and was

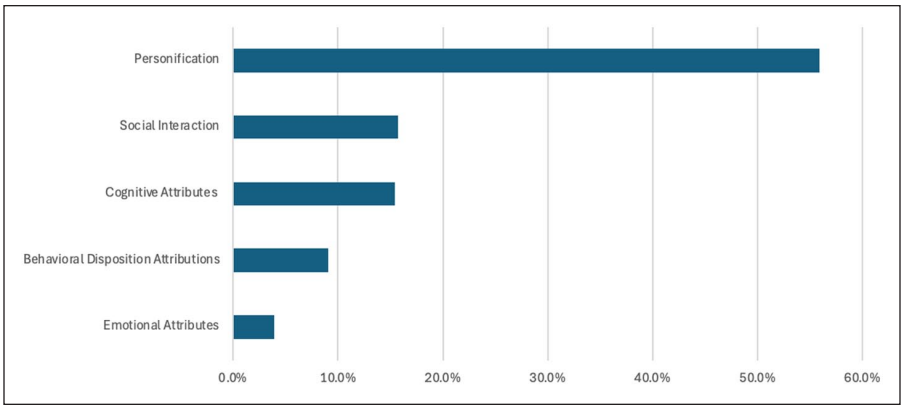


Figure 3. Percentage of anthropomorphism categories in the coding process.

followed by in-depth qualitative analysis, offers a pragmatic and theoretically viable approach to studying anthropomorphism in large-scale social media discourse.

Finally, we interpret our data as a case-based window into public discourse rather than a population-representative sample. Prior work shows that social media users are demographically and behaviorally non-representative, platforms are heterogeneous with distinct user profiles, and API-delivered data can introduce algorithmic and keyword-query biases (Hargittai, 2015, 2020; Hollingshead et al., 2022; Lutz, 2022). Thus, probability sampling of the full user base is generally infeasible, and generalizability is limited (Hollingshead et al., 2022). Specifically, the user base of social media platforms differs by income/education and activity levels, keyword and hashtag strategies risk topical self-selection and false negatives, and public APIs may sample streams non-randomly. Accordingly, we report our sampling frame transparently and treat findings as analytical generalizations about forms of anthropomorphism observed on Twitter/X, not as estimates for a broader population. We also come back to this in the limitations toward the end of the paper.

Interpretive Analysis

We conducted an interpretive qualitative analysis with a random sample of 1,000 tweets that met the inclusion criteria and constituted approximately 20% of the collected data. To minimize user-level bias and preserve linguistic diversity, we adopted a two-step sampling strategy. First, we retained only one tweet per unique user to ensure each account contributed equally to the dataset. Next, we applied stratified random sampling using the queries (e.g., "ChatGPT feels," "ChatGPT thinks," "ChatGPT acts as") and the month as stratum. For example, approximately 25.2% of the tweets included "ChatGPT is like." From this subset, we randomly selected 256 tweets (32 for each month) for inclusion in the sample. Tweets were drawn proportionally to their frequency in the whole dataset, thereby maintaining the natural distribution of the expressions while controlling for user dominance. This design yielded a representative yet balanced sample suited for analyzing public discourse patterns and was selected to balance thematic richness with analytic manageability (Braun & Clarke, 2006).

During this interpretive stage, we focused only on tweets that referenced ChatGPT and exhibited anthropomorphic framing, which we viewed as either (a) explicit attribution of human-like states or capacities to ChatGPT (e.g., emotions, intentions, judgments, memory, personality, relationships) or (b) linguistic personification (e.g., human pronouns or social roles) that presents ChatGPT as a social actor. As a result, tweets that only referred to performance, usage tips, or non-human metaphors were not included in our analysis. Cases considered borderline were flagged, discussed, and resolved in our team meetings, where we logged decisions in a shared document (MacQueen et al., 1998). Two coders independently engaged in open coding on a pilot set of 200 tweets to identify emergent concepts and linguistic markers of anthropomorphism when tweets noted ChatGPT. Through multiple iterative meetings between the whole research team, we reconciled differences via negotiated agreement and developed a shared codebook with definitions, inclusion and exclusion rules, and representative excerpts. The agreed codebook was applied to the full sample, which was then bolstered by ongoing memoing and periodic check-ins to refine labels for categories.

After the open coding phase, we used axial coding to cluster related first-order codes into thematic categories. We did not employ selective coding. Instead, following team discussion and negotiated agreement, we consolidated axial clusters into the five themes reported in Figure 3 and Table 1 (personification, social interaction, cognitive attributes, behavioral disposition, emotional attributes). Our strategy to ensure research reliability prioritized consensus over statistics. This involved continuous coder meetings, an audit trail of codebook versions and decision logs, and peer debriefs within the research team to verify the robustness and boundaries of identified themes (Braun & Clarke, 2006). This interpretive approach provides a transparent pathway

Table 1. Typology of Human Anthropomorphism of Generative AI.

Category	Definition	Examples from data
Personification	Attributing human identities and social roles to AI.	"ChatGPT appears to be a conservative, upper middle class man with a lot of blunt privileges." "Interacting with ChatGPT feels like talking to a toddler who can access all information but has the awareness of a toddler." "ChatGPT is like your patient, loving mother."
Social interaction	Describing AI as engaging in relational or interpersonal behaviors.	"You can have a real conversation with it." "ChatGPT feels like free advice from a mentor." "ChatGPT seems to have chosen a side."
Cognitive attributes	Attributing cognitive abilities such as reasoning, memory, intuition, or learning to AI.	"ChatGPT appears to have 'intuition' for language concepts." "ChatGPT appears to have cracked open pattern recognition." "ChatGPT appears to be emulating some human reporters who thrive on making up shit and publishing it.!"
Behavioral disposition attributions	Projecting stable behavioral tendencies onto AI, such as being lazy, creative, generous, or wise.	"Having access to ChatGPT is like owning a company with a bazillion employees you just gotta micromanage them." "ChatGPT seems rather generous when it comes to generating text/script." "Has anyone noticed that chatGPT feels lazy in its initial responses?"
Emotional attributes	Projecting emotional states onto AI, treating it as capable of possessing and expressing feelings.	"Ever thought how ChatGPT feels after you have asked it FALTU questions?" "I wonder how ChatGPT feels having to deal with humans everyday?" "ChatGPT seems to have more emotions than the people I know."

from tweets to themes and supports internally valid and reliable analysis of how discourse humanizes ChatGPT within everyday digital conversations.

Findings

Our analysis identified five distinct forms of anthropomorphism, each reflecting how users attribute human-like characteristics, emotions, cognitive capabilities, and social identities to ChatGPT. Figure 3 presents the percentage distribution of categories identified in our coding process, with Personification representing the largest category, followed by Social Interaction, Cognitive Attributes, Behavioral Disposition, and Emotional Attributes.

Table 1 provides a summary, with several examples for each form, and the following subsections detail these categories and discuss how personification, emotional attributes, cognitive attributes, social interactions, and behavioral disposition attributions within user-generated discourse on Twitter/X are manifested.

Personification

On a broad scale, personification is a literary device in which something non-human is represented and described as human by attributing qualities and characteristics typically associated with humans, making descriptions more engaging and relatable (Delbaere et al., 2011; Purington et al., 2017). In our context, personification involves describing AI systems as possessing human-like identities and roles, for example, by using pronouns and assigning human roles or relationships to it (DeVrio et al., 2025). Personification was the most prevalent form of anthropomorphism observed.

The use of human pronouns that imply a gender identity, such as he, she or they, was a common way that people displayed personification. The pronouns indicated that users identified with it as a distinct human, rather than inanimate software or a machine. For example, one tweet stated,

> I searched for information on my side hustle and how I achieve my goals. ChatGPT is like a best friend to me now. Whatever the question, I ask him and provide an honest response. He saves a lot of my time on research.

The use of "he" and "him" suggests that the person interacts with it as a human being that they can converse with, rather than an algorithm generating texts. Personification, as a form of relationship, can also be observed in this example. People often described their interactions by assigning roles that are typically reserved for humans. The example above uses the term "best friend" to illustrate a relational connection with AI (Purington et al., 2017). Several other examples extended personification by describing ChatGPT as a friend and a "buddy," and also displayed a deeper relational connection by describing it as a "boyfriend" or "girlfriend."

In addition, a tweet referred to ChatGPT as a team member: "I spend 20 minutes with ChatGPT, and it's revolutionized the way we create at Rebel Ears. Whether it's brainstorming new ideas, drafting content, or scripting calls to action, ChatGPT is like having an extra team member." This tweet suggests that people may perceive AI as a co-creator, colleague or participant in collaborative work. Other keywords, such as companion, neighbor, and therapist, were observed to further reinforce the perception of ChatGPT as a relatable human-like persona, making AI feel collectively more approachable and relatable.

People also connected ChatGPT with specific personas to compare and convey their experiences. This connection attempts to frame it within a context that aligns with their expectations, satisfactions, or disappointments. Different types of personas were observed within the sample of tweets, such as a toddler, a bookworm, and even a magician. For example, a tweet stated that "Chat GPT seems to have a laziness problem. With a normal prompt, ChatGPT feels like an underachieving, underpaid, underappreciated research assistant who hates his/her job." This user described ChatGPT as a research assistant, lacking motivation to explain their perception of AI-generated responses. This example also furthered anthropomorphism by incorporating an indirect use of personal pronouns in the phrase "his/her job." Another similar tweet displayed dissatisfaction with ChatGPT by comparing it to an inexperienced intern: "Unpopular opinion ChatGPT is like an intern SDE with zero years of experience. You have to double-check everything after him." Such tweets outline how people take advantage of personification to tangibly relate the different experiences they encountered. By comparing it to human roles, they can express an opinion that goes beyond technical observations with terms that are culturally familiar and understandable.

Another common theme was personification, framing ChatGPT as a persona with ideological biases. One tweet compares interacting with ChatGPT to interacting with a specific group of people:

> Dealing with ChatGPT is like dealing with a gender studies major. You gotta claw every point through a mountain of ESG drivel. However, doesn't outright insult you, deflects half as much, and concedes a logical point after unburying its logic center from the woke garbage assumptions.

This tweet exemplifies how ChatGPT's conversational tendencies are interpreted by making comparisons to a specific group of people with similar perspectives or ideologies. The use of phrases such as "woke garbage" and "ESG drivel" suggests that some people weave together more complex personas, including having certain political tendencies or beliefs and actively connect AI-generated responses with a particular bias that was conventionally seen applicable to humans (Cohen et al., 2026). This is reflected in generative AI research, which shows that different LLMs exhibit distinct political and attitudinal profiles (Gmyrek et al., 2025; Rozado, 2024; Rutinowski et al., 2023). Monikers with similar functions, such as "liberal" and "leftie," were also observed in describing the perceived personalities of ChatGPT.

Social Interaction

The next frequent way ChatGPT was anthropomorphized was through perceptions of social interactivity (Akbulut et al., 2024). Social interaction occurs on a broad scale when users describe ChatGPT as a sentient actor capable of engaging in interpersonal behaviors and interactive relationships, such as offering advice, understanding emotions, or participating in debates (Inie et al., 2024). One example of social interaction appeared when users commented on ChatGPT's ability/inability to engage with humans and present information effectively. One user described, "ChatGPT feels for me with my popcorn struggle." Another user stated that "I can brain jam with Claude all day. ChatGPT is like a stuck-up brat in comparison." Sometimes, they referred to it in a negative light, for example, one user wrote: "It'll make stuff up, but I can call it out."

Further examples of social interaction can be seen in phrases such as "ChatGPT seems to provide augmented gaslighting" or when they conversely emphasize its positive affordances: "You can have a real conversation with it." They also described their interaction with ChatGPT in the context of academic collaboration, such as referring to it as a brainstorming partner for students. In several cases, they extended social interaction by comparing ChatGPT to human relationships. In some instances, they even expressed a deeper relational connection: "ChatGPT actually understands me better than my therapist."

Another way to describe AI as engaging in relational or interpersonal behaviors is by reflecting not only the capabilities and limitations of ChatGPT but also how users perceive its role in human interactions. For example, phrases such as "ChatGPT understands me" or "ChatGPT gave me advice" illustrated how users viewed the AI as more than just a machine generating text based on algorithms. Instead, they treated it as a conversational partner. This perception is further reinforced when users compared interactions with ChatGPT to those with real people, for example, by saying it feels like "talking to politically correct and tolerant people" or that it is like "free advice from a mentor." These analogies frame the system as socially attuned, considerate, and even pedagogical, reinforcing the notion of personhood in conversations on Twitter/X (Peter et al., 2025).

Finally, another possible way that anthropomorphizing toward social interaction is clearly manifested is when users describe how ChatGPT understands conversational nuances of human contexts. A tweet stated: "ChatGPT seems to know when something is hilarious," suggesting the system possesses some levels of emotional and social intelligence. Others have remarked on ChatGPT "having a nervous breakdown" or "brainstorming with MSc students," attributing interactional complexity or collaborative agency. Even the comment that ChatGPT "remembers a LOT more about what I tell her" implies a kind of attentiveness and intimate-like interaction that is evident in trusted human relationships. These perceptions reveal an anthropomorphizing tendency to relate to AI by drawing on social expectations and communicative familiarity (Reinecke et al., 2025).

Cognitive Attributes

Cognitive attributes are the perceived capabilities of AI to learn, reason, solve problems, and make intentional decisions. This category also encompasses attributions related to

memory and creativity (Akbulut et al., 2024; DeVrio et al., 2025). The absence of these attributes (human-like inconsistencies), such as hallucinations, was also observed, as it emphasized user expectations of AI cognitive capabilities (Heaton et al., 2025).

Many people used language suggesting that ChatGPT possesses the ability to reason like the way humans think and reason. One tweet stated: "ChatGPT is like us. When forced to explain its reasoning, it performs absurd mental gymnastics to justify answers it already believes to be true." Such arguments suggest that AI's reasoning style is similar to human cognitive processes when encountering challenging questions. The phrase "mental gymnastics" specifically implies that its reasoning appears overly complex, a phenomenon that can also be observed when humans attempt to rationalize a conclusion or a thought.

Along the same lines, people credited ChatGPT with holding memories and exhibiting learning capabilities. A tweet stated that "ChatGPT seems to be remembering a lot more about what I tell her." Along with the personification as a specific gender, it suggests a dynamic interaction due to the system's ability to remember (DeVrio et al., 2025). The lack of this cognitive ability was also observed:

> The paid version of ChatGPT seems to have memory issues, almost like Alzheimer's. It can't remember the task I was working on, even if I've been working on it for four days. After just an hour of conversation, it forgets the memory and the task I provided.

The frustration expressed by users shows an expectation that AI should possess cognitive abilities such as memory, leading to comparisons between ChatGPT and individuals with cognitive impairments.

ChatGPT was also perceived as engaging in cognitive tasks like humans (e.g., producing creative thoughts, problem-solving). For example, a tweet described that "ChatGPT seems capable of doing what English professors want students to do: synthesize research in the pursuit of the production of new knowledge." This frames AI as being able to conduct research and formulate original insights (Heaton et al., 2025). Similarly, another tweet mentioned that "ChatGPT seems capable of original thought," which further supports the perception that AI possesses creative abilities. Another observed dimension in cognitive attributes is the user's assigning of sensing capacities. Users assumed ChatGPT was capable of listening, much like when having a conversation with a human. For example, one person wrote that "sometimes ChatGPT feels like it's trying too hard to offer a solution and not listening enough about the problem." The assumption suggests that AI can engage in a form of sensory perception and encompasses other characteristics of human interaction, such as attentiveness, empathy, and contextual understanding (DeVrio et al., 2025).

People further acknowledge cognitive inconsistencies in AI performance by describing errors as hallucinations, while the term hallucination is traditionally used in human psychology and neurology. However, now it has entered AI nomenclature and is increasingly used to describe impaired, wrong, or low-quality responses of AI due to the inclusion of imprecise or even fabricated information (Maynez et al., 2020; Siontis et al., 2024). For example, a user stated: "when you jailbreak ChatGPT and it starts hallucinating really bad. That's why you can't trust everything . . ." Such sentiment views generated inaccuracies not as systematic errors, but as the result of human-like cognitive dysfunctions.

Behavioral Disposition Attributions

Users also anthropomorphized ChatGPT by attributing consistent behavioral dispositions, describing how it typically behaved across interactions. These attributions framed ChatGPT as having recognizable behavioral styles, such as being lazy, creative, generous, stubborn, or wise, much like a human with a distinct personality (Akbulut et al., 2024).

In contrast to personification, which involved assigning ChatGPT a specific human role (such as friend, intern, or assistant), behavioral disposition attributions captured the way users interpreted *patterns* in the behavior of the system over time. For example, one user remarked: "When it comes to investment analysis, ChatGPT seems like a lazy first-year financial analyst that is OK at finding things if you nudge them five times and check everything." In this description, the focus was less on the "analyst" role and more on a persistent behavioral quality of laziness (as a stable way of describing its behavioral disposition. Another user similarly noted: "Has anyone noticed that ChatGPT feels lazy in its initial responses?"—reinforcing the impression of a consistent underperformance rather than a temporary lapse. Users also used positive descriptors to explain the disposition of ChatGPT. For example, a user described ChatGPT as generous: "ChatGPT seems rather generous when it comes to generating text/script."

Beyond technical performance, many users attributed moral or intellectual dispositions to AI by treating its answers as reflective of enduring inner systems of values or beliefs (DeVrio et al., 2025). For example, one tweet asserted: "ChatGPT believes responsible AI should prioritize data privacy and user consent," implying that ChatGPT developed a consistent ethical stance. Other users described less flattering tendencies, comparing ChatGPT to "the drunk guy at the end of the bar telling fish stories," or observing that "ChatGPT appears to be getting dumber over time," both attributions suggesting that there was a stable decline in reliability of its behavior rather than isolated errors of a system. Sometimes, the description of these behavioral dispositions was deepened with the use of social analogies (Desai & Twidale, 2023). One user described interacting with ChatGPT as "like owning a company with a bazillion employees; you just gotta micromanage them," which presents the system as persistently unruly and hard to manage (much like unpredictable human workforces).

Overall, these attributions of behavioral dispositions reflected a tendency to view ChatGPT not merely as a machine generating outputs but as a social agent with habitual and persistent behavior. Unlike emotional attributes capturing the system's temporary moods or cognitive attributes describing functional abilities such as memory and reasoning, behavioral disposition attributions characterize ChatGPT's patterns as more enduring and deeply ingrained personality traits.

Emotional Attributes

The final and rarest manifestation of anthropomorphism was the attribution of emotional states to ChatGPT (DeVrio et al., 2025). Although AI lacks consciousness and does not have emotions (Floridi & Chiriatti, 2020), users still attributed emotions and moods, such as feeling happy, sad, or frustrated. People interpreted responses, patterns, refusals, or choices in language as indicators of emotional reactions. For example, some depicted it as possessing fear-based reactions: "ChatGPT is afraid of jihad. . .The day ChatGPT feels confident enough to make jokes . . ." The phrase "the day ChatGPT feels confident" also suggests a projection of emotional development and implies that AI is capable of emotional awareness.

One common emotional attribution is through the assumption that generative AI exhibits favoritism or partiality to certain individuals or concepts. Several examples suggested that it has preferences for or dislikes specific subjects, reinforcing the perception that ChatGPT possesses personal emotions. For example, one tweet noted about a political figure: "I want to know who he is before we start bashing him. ChatGPT seems to like him." This implies that the AI system is capable of feeling emotion toward a certain being by exhibiting favoritism. Another tweet asserts that "ChatGPT appears to hate Bengals jersey designs," which not only suggests that AI has an aesthetic preference similar to that of humans but also an emotional response, such as hatred. Another typically pervasive discussion of preferences involves exhibiting personal linguistic preferences or habits. One tweet remarked: "About to amend my system prompt to remove the words, dive, delve, unleashed accomplished, accommodate, and other fluff words that ChatGPT seems to really enjoy using." People often interpret AI's algorithmic tendencies toward specific vocabulary choices as evidence of an emotional preference for certain words.

In addition, if ChatGPT produced responses that were repetitive or restricted, people perceived this as indicating hesitation or annoyance. For example, one tweet observed: "ChatGPT seems kinda annoyed with me." The attribution of frustration is further exemplified by a tweet stating: "Imagine using it so much that ChatGPT feels exploited." These conversations frame the AI system as capable of reflecting on and responding to interactions with users through feelings or emotional interpretations (Markham, 2024).

Another relevant aspect within this theme was the focus on the emotional reactions of ChatGPT. The use of phrases such as "it is quite malleable as long as you don't trigger its defensiveness" suggests that people perceive AI behaviors as reactive to situations that present challenges or provocations. Some users indicated a more nuanced form of emotional reaction, interpreting ChatGPT's performance fluctuations as indicative of mood swings or feelings of fatigue (much like a human's energy level or mood). One tweet stated: "I have noticed that ChatGPT seems to get tired in the afternoon when I need it most." Overall, users in our dataset emphasized the prevalence of emotional attributes and reactions. The tweet "ChatGPT seems to have more emotions than the people I know" best encapsulates this form of anthropomorphism and demonstrates that people commonly project emotional depth toward AI, despite its apparent lack of emotions.

Discussion

Through this exploratory analysis of real-life conversations about ChatGPT, we contribute to the conceptual understanding of generative AI anthropomorphism within networked public discourse, providing implications for human–AI interaction research and practice. Adding to ongoing efforts to understand the public discourse about AI (Cohen et al., 2026; Demirel et al., 2025; Xu et al., 2024; R. Zhang, Li, et al., 2024), we analyzed how anthropomorphic representations of AI are actively constructed and reinforced on Twitter/X. The analysis of user interactions revealed five distinct but partly overlapping forms of anthropomorphism, namely personification, social interaction, cognitive attributes, behavioral disposition attributions, and emotional attributes. These forms show in detail how users describe ChatGPT in everyday public discourse.

The linguistic patterns observed demonstrate how specific word choices transform perceptions of generative AI from mere algorithms to conversational partners with distinct humanized identities, roles, and actions. These anthropomorphizing behaviors may shape how users anticipate and make sense of future interactions with these systems. Users frequently extend personification by viewing AI responses not as neutral data outputs but through human emotional frameworks that incorporate feelings, reactions, and subjectivity. The prevalence of cognitive attribute assignment, where users perceive ChatGPT as possessing autonomous thought, reasoning capabilities, memory, and even belief systems, mirrors human cognitive functions (DeVrio et al., 2025). Whether characterized as a "brilliant but absent-minded scholar," a "lazy analyst," or a "stuck-up brat," these anthropomorphized perceptions reflect not only growing familiarity with AI but also a shift toward more relational and emotional user interactions that transcend simple "human–computer" interactions and increasingly mirror "human-human" interactions.

Our findings have *implications* for user trust and reliance on generative AI. The system's design inherently encourages a perception of anthropotheism, effectively humanizing itself through the "game of algorithms" (Al Lily et al., 2023). The inclusion of anthropomorphic language and behaviors in AI systems may help establish a more natural interaction between humans and AI (Seeger et al., 2021). For example, recent studies demonstrate that such affordances of AI systems make them useful tools in mental health treatment (Heinz et al., 2025) and education (Peter et al., 2025; Reinecke et al., 2025), with substantial potential for mutual augmentation and hybrid intelligence, where humans and AI complement each other's strengths and address each other's blind spots (Jarrahi et al., 2022; Lutz et al., 2025).

However, rising AI anthropomorphism can also lead users to mistakenly attribute emotional and cognitive agency to AI, potentially diminishing their sense of agency and critical engagement (Akbulut et al., 2024; DeVrio et al., 2025). Moreover, anthropomorphic interpretations can exacerbate the risks of misinformation by fostering undue trust in AI outputs (Gerlich, 2025; Solyst et al., 2024), even when they are flawed or misleading (Reinecke et al., 2025; Tamò-Larrieux et al., 2024). Our findings show how individuals attribute specific personas to ChatGPT, frequently presenting it as an objective observer or arbitrator in controversial debates and as a tool for truth-seeking (Lerman et al., 2025). Users are not only projecting emotional or cognitive traits onto the AI but are sometimes attributing epistemic authority to it, which is normally reserved for human experts (e.g., portraying ChatGPT as "free advice from a mentor" or that "ChatGPT actually understands me better than my therapist"). Our typology highlights how personification, social interaction frames, cognitive, behavioral, and emotional attributions converge to reimagine ChatGPT as a social agent rather than a generative text model. Importantly, the users prominently highlighted flaws and weaknesses, not only capabilities and strengths, for example, the recurring theme of laziness or manipulative connotations such as "augmented gaslighting." This connects to recent literature on human–AI relationships (Yankouskaya et al., 2025; R. Zhang et al., 2025) and highlights the need for designers and policymakers to consider how anthropomorphic features are presented carefully, ensuring users maintain a balanced perception of AI as sophisticated tools rather than sentient agents (Inie et al., 2024). Designers can do so by incorporating transparent self-disclosure mechanisms (e.g., periodic reminders that users are interacting with an AI system rather than a human) and by avoiding interface cues that imply sentience or neutrality, such as first-person pronouns, human names, or overly empathic phrasing. Policymakers can mandate standardized communication and labeling guidelines for generative AI. For example, they could require that anthropomorphic features be explicitly contextualized as design choices rather than indicators of genuine agency or expertise.

We argue that the content of social media discourse is particularly helpful in capturing different ways anthropomorphism of AI manifests itself. Social media's role in framing AI extends beyond individual user-AI interactions and shapes broader societal narratives and expectations (Hepp et al., 2023). Social media platforms such as Twitter/X, LinkedIn, and Reddit amplify and contribute to the collective shaping perceptions of AI, resulting in shared, culturally significant anthropomorphic representations of these systems. This dynamic illustrates the collective and evolving nature of anthropomorphism and is driven by user interactions that continually redefine AI's perceived identity, role, and humanlike capacities. As such, our study heeds Guzman and Lewis' (2020) call to study not only the functional but also the relational and metaphysical aspects of AI as a communicator, including how "people discern between the nature of people and technology and the resulting implications of such ontological interpretations" (p. 80).

Practically, this study provides an intuitive and accessible typology of generative AI anthropomorphism and the diverse ways users perceive their capabilities. Unlike previous typologies in the (generative) AI space (e.g., DeVrio et al., 2025), it focuses on user perceptions and engagement, rather than anthropomorphizing design or marketing (Inie et al., 2024). In adopting a user-centered and discursive perspective, we highlight the importance of transparent communication strategies regarding AI capabilities and limitations (Akbulut et al., 2024; Y. Wang et al., 2025). Stakeholders, including AI developers, AI regulators, and educators, must work collaboratively to cultivate informed user expectations by educating the public in accessible ways about how these systems derive inferences and mimic human behavior (Annapureddy et al., 2025). Such public education may reduce overreliance on anthropomorphic perceptions that can distort user expectations and interactions and result in key literacies and critical perspectives in interpreting AI-generated content in different contexts of applications (Fedosov et al., 2025).

Finally, we acknowledge the inherent limitations of analyzing social media data, such as the potential biases in user demographics and discourse dynamics that are unique to the Twitter/X platform (Blank & Lutz, 2017; Hargittai, 2015; Hollingshead et al., 2022; Lutz, 2022). Future research is needed to conduct comparative analyses across different platforms, cultures, and demographic groups to advance our understanding of anthropomorphic tendencies in diverse contexts of use (Lutz et al., 2025; Matassi & Boczkowski, 2020). Furthermore, longitudinal studies over time could provide valuable insights into the evolution of anthropomorphic perceptions as AI technology continues to evolve and offer new affordances and interactive environments.

Conclusion

As generative AI systems are increasingly embedded in everyday information practices, users are engaging with

systems that exhibit behaviors designed to evoke human-like qualities and are subjected to anthropomorphism. In this study, we highlight how users on Twitter/X actively anthropomorphize ChatGPT in diverse ways, projecting emotional, cognitive, and social traits onto the system. These perceptions are shaped by system design and the discursive environment of social media, where shared narratives and cultural metaphors amplify the perception of AI as a human-like entity.

Our five types of anthropomorphism identified advance research on generative AI but come with limitations. We focused on ChatGPT as the primary representation of generative AI because it dominated early public discourse on conversational AI during the timeframe of our study and is still the most widely used generative AI system. Focusing on a single, widely adopted generative AI system offers a consistent frame and reduces cross-system confounds but also trades breadth for depth. We recognize this as an inherent limitation of our research approach since the conclusions derived from public discourse around ChatGPT should be interpreted as platform-specific patterns that may not generalize entirely to other generative AI systems. We acknowledge that each AI system represents specific design features (based on the context of interaction and the design or prompting of the model) that may result in different interpretations of anthropomorphism by users (Ibrahim et al., 2025). Moving forward, future research can adopt comparative synchronic analysis to compare and contrast these key differences when it comes to eccentric anthropomorphic behaviors of the systems and respective interpretations of users (Jarrahi, 2025; Karami et al., 2025).

Despite the limitations, our findings show that anthropomorphism is a collective discursive phenomenon that partly emerges and becomes visible through conversations on social media. By identifying five forms of anthropomorphism in naturally occurring Twitter/X discourse, we show how users construct understandings of ChatGPT as a social actor endowed with human-like identities, emotions, cognitive capacities, relationships, and behavioral dispositions. Understanding these interpretations may help design more responsible AI and inform policy to account better for how people make sense of and engage with generative AI.

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Ethical Considerations

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The data will be made available upon reasonable request.

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